



NATIONAL INSTITUTE OF TECHNOLOGY PATNA
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
MID Semester Examination Jan - June 2023

B. Tech: CSE IVth semester
Course Code: CS44115
Maximum Time: 2 Hours

Course Name: **Design and Analysis Of Algorithm**
Max. Marks: 30

Instructions:

1. Attempt all the questions.
2. Paper is printed on both sides.
3. Assume any suitable data, if necessary.
4. Answer all parts of the question at the same place.

Question	CO	Marks
Q.1a) What is the time complexity of the following function fun1()? Explain your answer. <pre>int fun1(int n) { int i, j, s=0; for (i = n; i >= n; i /= 2) for (j = 0; j < i; j++) s += 1; return s; }</pre>	CO1	2
Q.1b) What is the running time of QUICKSORT when all elements of array A have the same value? Explain your answer.	CO2	2
Q.1c) Consider the following code fragment. <pre>int a[] = {2, 1, 4, 3, 5, 6, 7, 8, 9, 0}; int fun(int b[], int n) { if (n==1) return b[n-1]; else return b[n] + 2*fun(b, n-1); }</pre> What is the result of the function call fun(a, 3)? Also, compute the asymptotic running time of this function in terms of n?	CO1	3
Q.1d) What is the solution to the recurrence $T(n) = 2T(\sqrt{n}) + n$, $T(1)=1$. Show the steps.	CO2	3
Q2. Given a set S of n integers and another integer x, determine whether or not there exist two elements in S whose sum is exactly x. Describe a $\Theta(n \log n)$ time algorithm for the above problem.	CO2	5
Q3. Design the Huffman code for the following set of frequencies which is based on the first 8 Fibonacci numbers. Explain each step. a: 1 b:1 c:2 d:3 e:5 f:8 g:13 h:21 Can you generalize your answer to find the optimal code when the frequencies are the first n Fibonacci numbers? What will be the codeword for the nth most frequent letter? What will be the codeword for $k < n^{\text{th}}$ most frequent letter?	CO4	3+2

Q4. Mr. Sharma once buys different amounts (in Units) of precious metals (like Gold, Silver, Platinum, etc.), as given in the Table below. He is currently in urgent need of Rs 800K. So, he decided to sell some units (a fraction of the unit can be sold) of different metals to get this amount (Rs. 800K). Now, he wants to sell only those metals which are making the maximum profit percentage.

Find the quantity of different metals to be sold by Mr. Sharma to maximize the profit percentage subject to generate the above-said amount.

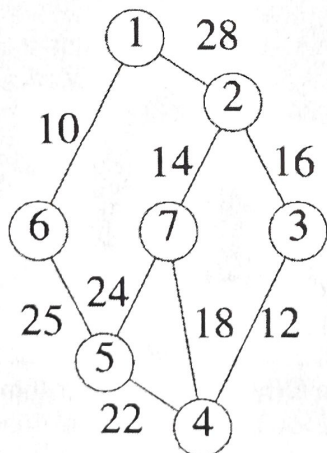
Also, write an algorithm for Mr. Sharma to decide units of metals to be sold to make maximum profit percentage and generate a specific amount.

Table: Mr. Sharma holdings of Different Metals

Precious Metal Name	No. of Units	Per Unit Purchasing Price in Rs	Current Per Unit Selling Price in Rs
Gold	5	81K	90K
Silver	8	35K	40K
Platinum	2	60K	80K
Rhodium	4	50K	55K
Ruthenium	5	27K	30K
Iridium	3	25K	30K
Osmium	6	30K	45K
Rhenium	2	18K	21K
Gallium	20	8K	10K

CO4 5

Q5.) For the below-given graph, find a Minimum Spanning Tree through Prim's algorithm by taking '1' as the starting node. Show all the stages in the diagrammatic form only. Also, perform the complexity analysis of prim's algorithm.



CO4 5